

4 Homeostasis is the state maintained by living systems.

(a) Feedback mechanisms are involved in maintaining homeostasis.

Describe the difference between a negative feedback mechanism and a positive feedback mechanism.

(2)

(b) Which statement about the homeostasis of thermoregulation when the human body temperature is 37°C is correct?

(1)

- ☐ **A** thermoregulation is no longer required
- ☐ **B** the body no longer generates any heat
- ☐ **C** the body will be generating more heat than it can lose
- ☐ **D** the body temperature is maintained in a dynamic equilibrium

- (c) The health of a kidney can be determined by measuring the glomerular filtration rate (GFR).

This method measures the rate at which a small molecule called creatinine is filtered from the blood.

- (i) Explain how small molecules like creatinine are filtered from the blood in the glomerulus.

(3)

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- (ii) In 15 hours, a person excreted 1960 mg of creatinine in the urine.

Calculate the GFR for this person.

Give your answer to 3 significant figures.

(1)

Answer

(d) In type II diabetes, body cells take up less glucose in response to insulin.

Many people with type II diabetes are overweight or obese.

Some of these individuals have been given surgery to help weight loss.

A clinical trial investigated the effect of surgery on the uptake of glucose by body cells in the presence of insulin.

This trial used groups of people who were non-diabetic, people in the early stage of diabetes (pre-diabetic) and people who were type II diabetics.

Group	Mean uptake of glucose / a.u.	
	Before surgery	After surgery
Non-diabetic	1.07 ± 0.29	1.39 ± 0.58
Pre-diabetic	0.55 ± 0.32	1.30 ± 0.88
Type II diabetic	0.40 ± 0.24	1.10 ± 0.87

(i) Comment on the results of this trial.

(3)

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- (ii) Calculate the difference that surgery makes to the mean uptake of glucose in the non-diabetic group and the type II diabetic group.

(2)

Answer

(Total for Question 4 = 12 marks)

